

AUTOMATIC OPTICAL INSPECTION: A Game Changer For PCB Manufacture

Increase in the complexity of electronic boards shows manual inspection may not be a viable option. With the marketplace now requiring high quality, no-error products, there is a strict need for reliable fast methods. One of the effective solutions is to integrate AOI in your electronics test strategy

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Automatic optical inspection (AOI) is a key technique used in manufacturing and testing of electronics printed circuit boards (PCBs). It enables fast and accurate inspection of electronic assemblies, thus ensuring production of high quality of boards without defects. This technique usually comes at the stage just after the soldering process to find and fix faults, especially in the solder and assembly area. In this way of rapid response, defects are recognised quickly and rectified before too many boards are built with the same errors.

Not only identifying errors and pausing production of defective boards, AOI provides data needed to improve the performance of the entire SMT process.

AOI uses various techniques like template matching or pattern matching to check the quality of a board. It compares the captured image with the image of a 'golden board'—an ideal board that is of acceptable quality and standard, stored in the database. It can be used to adjust and improve the quality of manufacturing without stopping the process. The need for a dynamic SMT inspection equipment has driven manufacturers to connect their equipment and process data to enable smart, adaptable functioning of AOI machines.

What an AOI machine does

These inspection systems use visual methods to monitor PCBs to detect various surface defects like nodules, scratches, and

stains and other dimensional defects by in-circuit testing (ICT), such as open circuits, shorts, discontinuity, and thinning of the solder. They can also detect incorrect components, missing components, and incorrectly placed components.

An AOI machine uses optical light energy to capture and analyse images of PCBs to verify the assembled electronic components, including presence and position, soldering (solder joints and bridges), polarity, marking (like inscription, signs, colour), flux deposits, microcracks, and subsurface defects. It performs with the help of illumination, CCD image system and processing software, machine vision

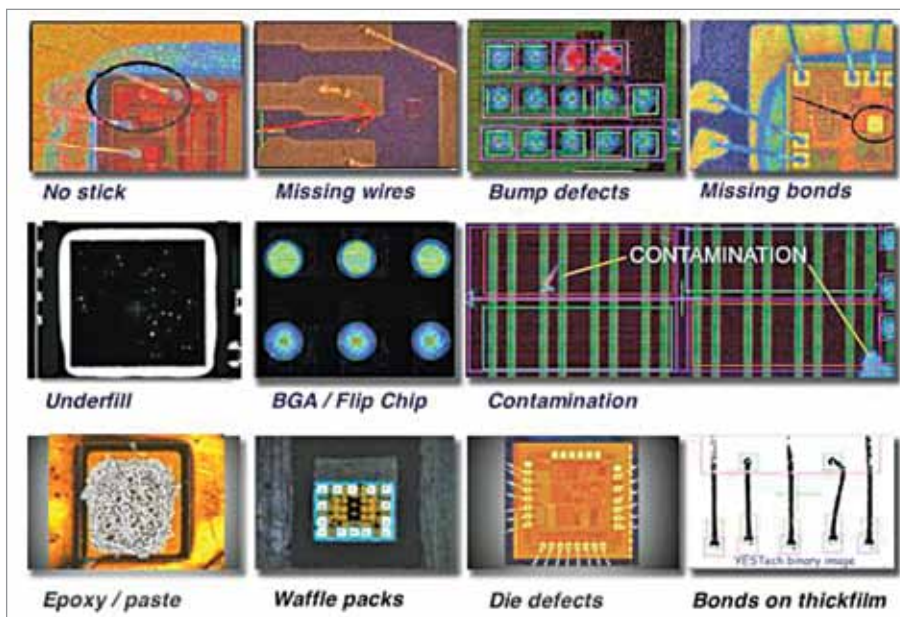


Fig. 1: Optical Inspection (Source: Nordson YESTECH)